KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 202: FLUID MECHANICS I

2ND SEMESTER 2020/2021 **TIME: 2 HOURS**

INSTRUCTIONS:

i) THIS PAPER HAS A TOTAL OF FIVE QUESTIONS.

ii) ATTEMPT QUESTION 1 AND ANY OTHER THREE (3) QUESTIONS.

QUESTION 1 – COMPULSORY (19 Marks)

a. State Newton's Law of viscosity. [1]

b. Differentiate between dynamic viscosity and kinematic viscosity. State their units of measurements. [3]

$\tau = u\frac{dy}{dy}$ $u = \frac{\tau}{m} \frac{}{dy}$ N/m^2 L dmks $\frac{m/s}{m}$ m^2/s

c. The space between two square flat parallel plates is filled with oil. Each side of the plate is 60 cm. The thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 m/s requires a force of 98.1 N to maintain the speed. Determine: $\frac{dy}{dy}$

   i. The dynamic viscosity of the oil in poise, and [4]

   ii. The kinematic viscosity of the oil in stokes if the specific gravity of the oil is 0.95 [3]

   950 x

d. Define the following terms: (i) Surface Tension and, (ii) Capillarity. [2]

e. The surface tension of water in contact with air at 20°C is 0.0725 N/m. The pressure inside a droplet of water is to be 0.02 N/cm^2 greater than the outside pressure. Calculate the diameter of the droplet of water. [3]

f. Find out the minimum size of glass tube that can be used to measure water level if the capillary rise in the tube is to be restricted to 2 mm. Consider surface tension for water in contact with air as $\sigma = 0.073575\ N/m$. [3]

LT^{-1}

QUESTION 2 (17 Marks)

a. State the Pascal's law. [1]

b. A hydraulic press has a ram of 20 cm diameter and a plunger of 3 cm diameter. It is used for lifting a weight of 30 kN. Find the force required at the plunger. [4]

c. Illustrate the relationship between pressures by use of a clearly labelled diagram. [3]

d. Differentiate between *simple manometers* and *differential manometers.* [2]

e. A differential manometer is connected at the two points A and B as shown in **Fig. Q2.1**. At B, air pressure is 9.81 N/cm^2 (abs), find the absolute pressure at A in **kPa**. [7]

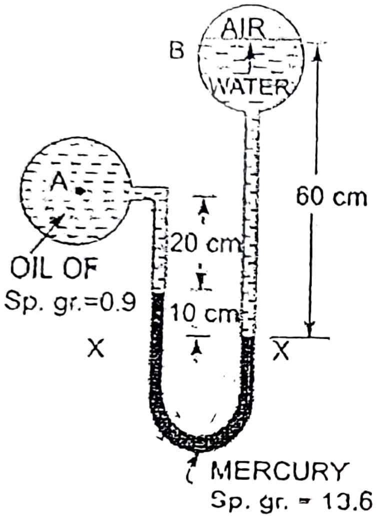


Figure Q2.1

Absolute pressure = atmospheric pressure - guage pressure

13600x76

QUESTION 3 (17 Marks)

a. Define the terms (i) Total Pressure and, (ii) Centre of Pressure. [2]

b. Determine the total pressure and depth of centre of pressure on a plane rectangular surface of 1 m wide and 3 m deep when its upper edge is horizontal and, (i) coincides with water surface, (ii) 2 m below the free water surface.

[5]

c. The end gates ABC of a lock are 8 m high and when closed make and angle of 120°. The width of lock is 10 m. Each gate is supported by two hinges located at 1 m and 5 m above the bottom of the lock. The depth of water on the upstream and downstream sides of the lock are 6 m and 4 m, respectively. Find;

   i. Resultant water force on each gate [4]

   ii. Reaction between the gates AB and BC [2]

   iii. Force on each hinge, considering the reaction of the gate acting in the same horizontal plane as resultant water pressure. [4]

QUESTION 4 (17 Marks)

a. Define the following terms;

i. Buoyancy [1]

ii. Centre of buoyancy [1]

iii. Metacentre [1]

iv. Meta-centric height [1]

b. A wooden block of width 2 m, depth 1.5 m and length 4 m floats horizontally in water. Find the volume of water displaced and position of centre of buoyancy. The specific gravity of the wooden block is 0.7.

[4]

c. A body of dimensions $1.5\,m \times 1.0\,m \times 2\,m$, weighs 1,962 N in water. Find its weight in air. What will be its specific gravity? [4]

d. A rectangular pontoon is 5 m long, 3 m wide and 1.20 m high. The depth of immersion of the pontoon is 0.80 m in sea water. If the centre of gravity is 0.6 m above the bottom of the pontoon, determine the meta-centric height. Take the density of sea water as 1,025 kg/m^3.

[5]

$\frac{IG}{A\bar{h}} + A\bar{h}$ $0.8 \times 5 \times 3$ $I = \frac{bd^3}{12}$

$GM = \frac{I}{V} - BG$

$I = \frac{bd^3}{12} = \frac{5 \times 3^3}{12} = 11.25$ m

$V = 1.2 \times 3 \times 5 = 18$

G 0.6, B 0.4, 1.2

$BG = 0.6 - 0.4 = 0.2$

$GM = \frac{11.25}{18} - 0.2 = 0.425$ m

Bh^3

QUESTION 5 (17 Marks)

a. Distinguish between the following terms:

i. Steady and unsteady flow [2]

ii. Laminar and turbulent flow [2]

iii. Free vortex and forced vortex flow. [2]

b. A 40 cm diameter pipe, conveying water, branches into two pipes of diameters 30 cm and 20 cm, respectively. If the average velocity in the 40 cm diameter is 3 m/s, find the discharge in this pipe. Also determine the velocity in 20 cm pipe if the average velocity in 30 cm diameter pipe is 2 m/s. [4]

c. An open circular cylinder of 20 cm diameter and 100 cm long contains water upto a height of 80 cm. It is rotated about its vertical axis. Find the speed of rotation when; (i) no water spills, and (ii) axial depth is zero. [7]

*****************************END*****************************

4

PART II

Question 4

(a) Derive the formula for actual flow through a triangular notch taking C_d to be 0.62

(b) A weir is 0.9m long and is situated centrally in a channel 1.2m wide. If the head above the sill is 25cm, calculate the actual discharge given C_d is 0.6.

Question 5

(a) What is a manometer? Using a simple sketch discuss the functionalities of a differential manometer.

(b) A U-tube contains water and oil. The oil of density $800 kg/m^3$, rest on the surface of the water in the right-hand leg to a depth of 5cm. if the level of water in the left hand leg is 10cm above the level of water in the right hand leg, determine the pressure difference between the two legs. The density of water is $1000 kg/m^3$.

Question 6

A rectangular pontoon has a width of 6.5m, a length L of 11.5m and a draught D of 1.7m in fresh water. Calculate

(i) The weight of the pontoon

(ii) The draught in sea water $\rho = 1025 kg/m^3$

(iii) The max load that can be carried by the pontoon in sea water if the max allowable draught is 1.9m

(15 Marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING/ BACHELOR OF SCIENCE IN AGRICULTURAL AND BIOS STEMS ENGINEERING

ECV 202/ EBE 202: FLUID MECHANICS I

DATE: FRIDAY 15TH MARCH 2019 TIME: 2.00 P.M. – 4.00 P.M.

INSTRUCTIONS

I. Attempt QUESTION ONE and any other Two questions
II. Reasonable assumptions may be made where necessary

Question 1 Ns/m²

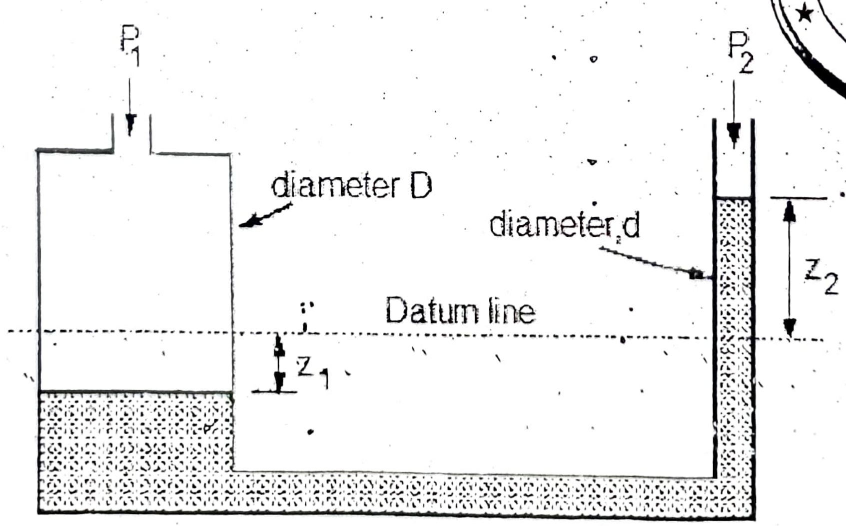
(a) Show that kinematic viscosity has primary dimension L^2T^{-1}. (3Marks)

(b) State the Bernoulli's equation and the four assumptions governing its application. (6 Marks)

(c) State the six forces influencing fluid motion. (6 Marks)

(d) For the U-tube manometer shown below, prove that the difference in pressure is given by:

$$P_1 - P_2 = \rho g Z_2 \left[1 + \left(\frac{d}{D}\right)^2\right]$$



(5Marks)

(e) How deep can a diver descend in the ocean water without damaging his watch, which will withstand an absolute pressure of 5.5bar? Take the density of ocean water, $\rho = 1025kg/m^3$ (4Marks)

(f) Using a relevant sketch demonstrate that the actual discharge through an orifice is $Q_{Act} = AC_d\sqrt{2gh}$ (6 marks)

Question 2

(a) A hydraulic jack having a ram 200 mm in diameter lifts a weight W = 40 kN under the action of a 25 mm plunger. What force is required on the plunger to lift the weight? (5Marks)

(b) A flat rectangular plate, dimension 1.5m length by 0.5m width is immersed in saline water (density 1025kg/m³) such that their greatest and least depths are 1.50m and 0.60m respectively. Determine the force exerted on one face by the water pressure. (9Marks)

(c) In a fluid the velocity measured at a distance of 75mm from the boundary is 1.125m/s. the fluid has absolute viscosity 0.05 Pa s and relative density 0.93. What is the velocity gradient and shear stress at the boundary assuming a linear velocity distribution? Calculate its kinematic viscosity. (6 Marks)

Question 3

(a) Using a sketch and the relevant equations, show that the buoyancy or upthrust force on a floating body, $F = \rho g V_i$ (5 Marks)

(b) A rectangular pontoon has a width of 6m, a length L of 12m and a draught D of 1.5m in fresh water. Calculate

(i) The weight of the pontoon

(ii) The draught in sea water $\rho = 1025kg/m^3$

(iii) The max load that can be carried by the pontoon in sea water if the max allowable draught is 2.0m (15 Marks)

Question 4

(a) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 4mm of water, calculate the air velocity assuming the coefficient of the pitot-tube to be unity. Density of air = 1.2KG/M^3.

(10 marks)

(b) A venturi meter having an entrance diameter of 300mm and a throat diameter of 200mm is used to measure the volume of oil flowing through a pipe. The coefficient of discharge Cd is 0.95. If the density of oil ~~and~~ is equal to 0.9kg/m^3. Making reasonable assumptions, calculate the volume flow per second when the pressure difference between the entrance and the throat as measured on a water U-tube is 63mm. (10 Marks)

Question 5

(a) A tank has a circular orifice 20mm diameter in the vertical side near the bottom. It contains water to a depth of 1m above the orifice with oil of relative density 0.8 for a depth of 1m above the water. Acting on the upper surface of the oil is an air pressure on 20kN/m^2. The jet of water issuing from the orifice travels a horizontal distance of 1.5m from the orifice while falling a vertical distance of 0.156m. If the coefficient of contraction of the orifice is 0.65, estimate, find:

(i) The value of the coefficient of velocity, and

(ii) The actual discharge through the orifice. (10 Marks)

(b) Water flow over a rectangular notch and then over a 120^0 vee notch. The rectangular notch is 700mm long and the depth of water is 120mm. Find the corresponding depth over the V-notch. Assume coefficients of discharge to be 0.62 and 0.60 for the rectangular and V-notch respectively. (10 Marks)

CONTINUOUS ASSESSMENT TEST 2018/2019

BACHELOR OF SCIENCE IN CIVIL/ AGRICULTURAL & BIOSYSTEMS ENGINEERING

ECV 202 / EBE 202: FLUID MECHANICS I

Date: 23rd. Jan 2019: 7.30 - 9.30am

(i) ATTEMPT ALL THREE QUESTIONS

(ii) REASONABLE ASSUMPTIONS MAY BE MADE WHERE NECESSARY

Question 1

(a) What is the definition of fluid in fluid mechanics

(b) A hydraulic jack having a ram 250 mm in diameter lifts a weight W = 40 kN under the action of a 35 mm plunger. What force is required on the plunger to lift the weight? $F = W/A$

(c) Show that dynamic viscosity has primary dimension $ML^{-1}T^{-1}$.

(d) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 5mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be 0.98. Density of air = $1.2 Kg/m^3$.

Question 2

(a) Using a simple sketch state and proof the Parallel Axis Theorem

(b) A flat rectangular plate, dimension 1.5m length by 0.5m width is immersed in saline water (density $1025 kg/m^3$) such that their greatest and least depths are 1.80m and 0.80m respectively. Determine the force exerted on one face by the water pressure.

Question 3

Crude oil of density $850 Kg/m^3$ flows upwards at a volume rate of $0.06\ m^3/s$ through a vertical venturi meter which has an inlet diameter of 200mm and a throat diameter 100mm. The venturi coefficient is 0.9. The vertical distance between the pressure tappings is 300mm and they are connected:

(i) To two pressure gauges calibrated in kN/m^2 and positioned at the levels of their corresponding tapping points.

(ii) To the two limbs of a mercury manometer ($\rho_{Hg} = 13600 Kg/m^3$), the connecting tube being filled with oil.

Determine the difference of the readings of 2 pressure gauges and difference of the level of mercury column in the manometer.

CONTINUOUS ASSESSMENT TEST 2018/2019

BACHELOR OF SCIENCE IN CIVIL/ AGRICULTURAL & BIOSYSTEMS ENGINEERING

ECV 202 / EBE 202: FLUID MECHANICS I

(i) ATTEMPT A TOTAL OF FOUR QUESTIONS

(ii) TWO QUESTIONS SHOULD BE FROM PART I AND TWO FROM PART II

DATE: 27/02/2019

PART I

Question 1

(a) Derive the theoretical flow equation for a pitot static tube.

(b) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 3.0mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be 0.95. Density of air = 1.2Kg/m^3.

Question 2

(a) State the Pascal Law and using an example cite its engineering application.

(b) A venturi meter having a throat 110mm in diameter is fitted in a pipeline 220mm in diameter through which oil of specific gravity 0.9 is flowing at the rate of 0.15m^3/s. The inlet and throat of the meter are connected differentially to a U-tube manometer containing mercury of specific gravity 13.6 with oil immediately above this. Starting from Bernoulli's equation find the coefficient of discharge for the meter if the difference in mercury levels is 0.63m.

Question 3

(a) A tank 2.8 m high standing on the ground is kept full of water. There is an orifice in its vertical side at a depth hm below the surface of water. Find the value of h in order that the jet may strike the ground at a maximum distance from the tank.

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING/ BACHELOR OF SCIENCE IN AGRICULTURAL AND BIOS STEMS ENGINEERING

fluid mechanics
Kinetic Viscosity

ECV 202/ EBE 202: FLUID MECHANICS I

DATE: FRIDAY 15TH MARCH 2019 TIME: 2.00 P.M. – 4.00 P.M.

theoretical flow

INSTRUCTIONS

I. Attempt QUESTION ONE and any other Two questions
II. Reasonable assumptions may be made where necessary

Question 1

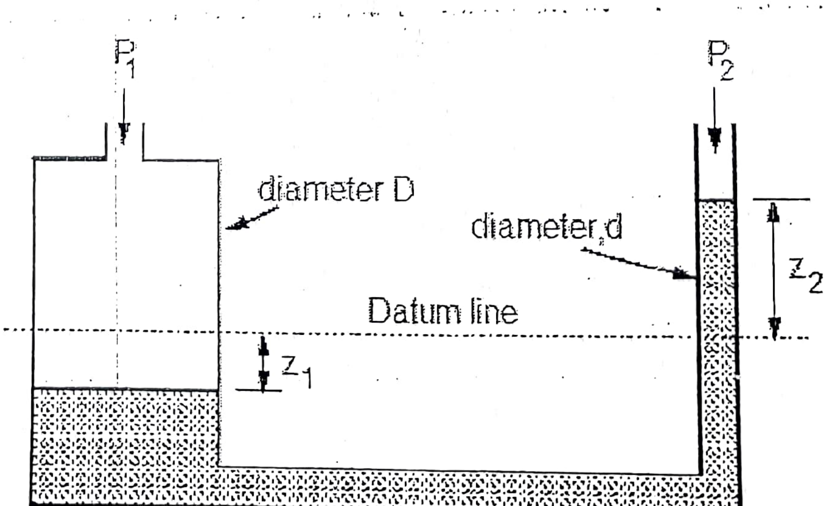
(a) Show that kinematic viscosity has primary dimension L^2T^{-1}. (3Marks)

(b) State the Bernoulli's equation and the four assumptions governing its application. (6 Marks)

(c) State the six forces influencing fluid motion. (6 Marks)

(d) For the U-tube manometer shown below, prove that the difference in pressure is given by:

$$P_1 - P_2 = \rho g Z_2 \left[1 + \left(\frac{d}{D}\right)^2\right]$$



(5Marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

(e) How deep can a diver descend in the ocean water without damaging his watch, which will withstand an absolute pressure of 5.5bar? Take the density of ocean water, $\rho = 1025 kg/m^3$ (4Marks)

(f) Using a relevant sketch demonstrate that the actual discharge through an orifice is $Q_{Act} = AC_d\sqrt{2gh}$ (6 marks)

Question 2

(a) A hydraulic jack having a ram 200 mm in diameter lifts a weight W = 40 kN under the action of a 25 mm plunger. What force is required on the plunger to lift the weight? (5Marks)

(b) A flat rectangular plate, dimension 1.5m length by 0.5m width is immersed in saline water (density $1025 kg/m^3$) such that their greatest and least depths are 1.50m and 0.60m respectively. Determine the force exerted on one face by the water pressure. (9Marks)

(c) In a fluid the velocity measured at a distance of 75mm from the boundary is 1.125m/s. the fluid has absolute viscosity 0.05 Pa s and relative density 0.93. What is the velocity gradient and shear stress at the boundary assuming a linear velocity distribution? Calculate its kinematic viscosity. (6 Marks)

Question 3

(a) Using a sketch and the relevant equations, show that the buoyancy or upthrust force on a floating body, $F = \rho g V_i$ (5 Marks)

(b) A rectangular pontoon has a width of 6m, a length L of 12m and a draught D of 1.5m in fresh water. Calculate

(i) The weight of the pontoon

(ii) The draught in sea water $\rho = 1025 kg/m^3$

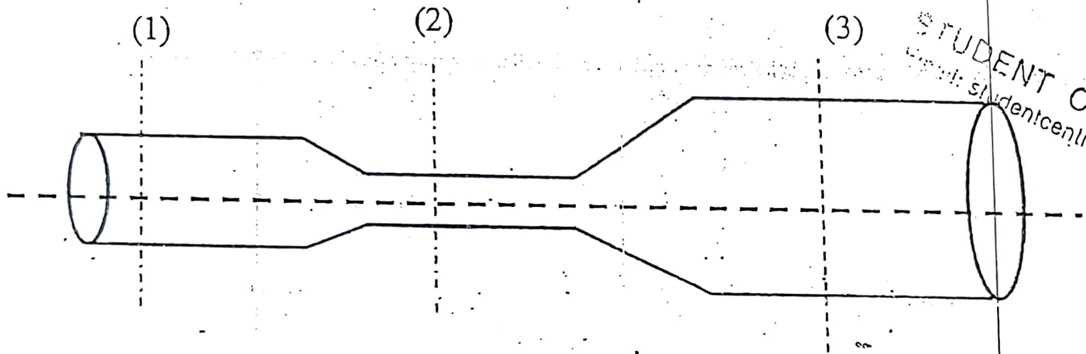
(iii) The max load that can be carried by the pontoon in sea water if the max allowable draught is 2.0m (15 Marks)

ECV 202 11/12/2014

(b) If for two dimensional flow, the stream function is given by $\psi = y(3x^2 - y^2)$, obtain the velocity components and velocity at point (2,1). (8 marks)

5. (a) State the Bernoulli's theorem for a liquid (3 marks)

(b) An incompressible fluid flows through the circular pipe as shown in the figure below at the rate of Qm^3/s, it is assumed that the velocities at section (1), (2) and (3) are uniform, what are the velocities given that the diameter of pipe at three sections are $d_1 = 0.4m$, $d_2 = 0.2m$ and $d_3 = 0.75m$ respectively and Q = 0.3m3/s. (7 marks)



(a) A vessel containing a liquid of mass density $930kg/m^3$ is given a constant vertical acceleration $f = 4.8m/s^2$ in an upward direction. If the vessel is 1.2m wide and 1.5m in breadth and the depth of the liquid is 0.9m, calculate the force on the bottom of the vessel;

i. While it is being accelerated (6 marks)

ii. When the acceleration ceases and the vessel continues to move at a constant velocity of 6m/s vertically upward. (4 marks)

UNIVERSITY EXAMINATIONS

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECV 202: FLUID MECHANICS I

DATE: THURSDAY 11TH DECEMBER, 2014 **TIME: 11.00 A.M – 1.00 P.M**

INSTRUCTIONS:

- *Answer question 1 and any other TWO questions*

1. (a) (i) At a certain point in castor oil the shear stress is $0.216 N/m^2$ and the velocity gradient is $0.216 s^{-1}$. If the mass density of castor oil is $959.42 kg/m^3$. Find the kinematic viscosity (2 marks)

(ii) If $6m^3$ of oil weighs 47KN, calculate the specific weight, mass density and specific gravity. (3 marks)

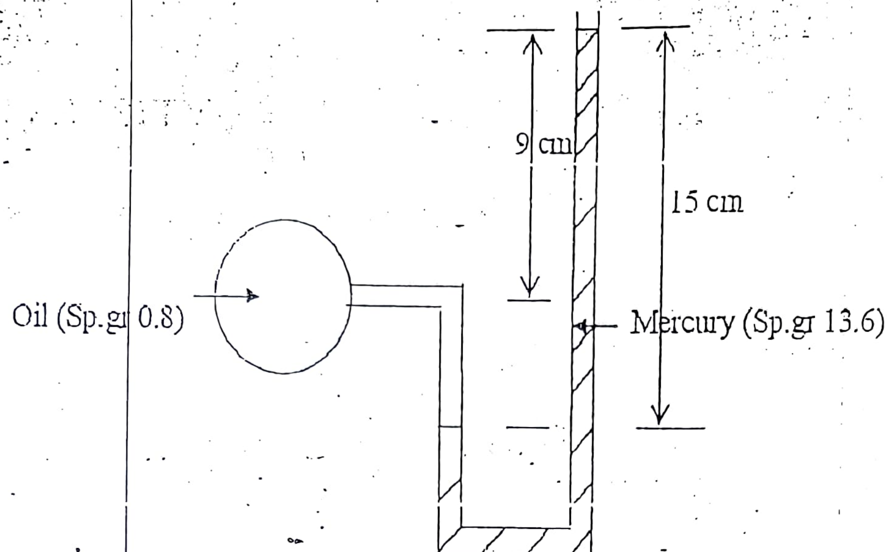
(iii) A clean tube of internal diameter 3 mm is immersed in a liquid with coefficient of surface tension 0.48 N/m. The angle of contact of the liquid with the glass can be assumed to be 130^0. The density of the liquid is $13600 kg/m^3$. What would be the level of the liquid in the tube relative to the free surface of the liquid outside the tube? (3 marks)

(iv) What is the bulk modulus of elasticity of a liquid which is compressed in a cylinder from volume of $0.0125 m^3$ at $80 N/cm^2$ to a volume of $0.0124 m^3$ at $150 N/cm^2$. (2 marks)

(b) Show that the intensity of pressure at a point in a fluid at rest is the same in all directions (6 marks)

(c) A square surface 3 m x 3m lies in a vertical plane. Determine the position of the centre of pressure and the total force on the square, when its upper edge is (a) in water surface and (b) 15m below the water surface. (4 marks)

(d) A U tube manometer is used to measure the pressure of oil (sp. gr. 0.8) flowing in a pipeline. Its right limb is open to the atmosphere and the left limb is connected to the pipe. The centre of the pipe is 9cm below the level of mercury (sp. gr 13.6) in the right limb. If the difference of mercury level in the two limbs is 15cm, determine the absolute pressure of the oil in the pipe in KPa. (8 marks)



2. (a) A plane surface of area A is totally immersed in a liquid of specific weighγ. If the surface is inclined at an angle θ to the horizontal and its centroid is at a vertical depth $\bar{y}$ below the free surface, derive an expression for the resultant force F acting on the inclined plane surface. (9 marks)

   (b) A 3.6m by 1.5m wide rectangular gate MN is vertical and is hinged at appoint 15cm below the centre of gravity of the gate. The total depth of water is 6m. What horizontal force must be applied at the bottom of the gate to keep the gate gate closed. (5 marks)

   (c) A triangular gate which has a base of 1.5m and an altitude of 2m lies in a vertical plane. The vertex of the gate is 1m below the surface of a tank which contains oil of specific gravity 0.8. Find the force exerted by the oil on the gate and the position of the centre of pressure. (6 marks)

3. (a) Show that if B is the centre of buoyancy and M is the metacentre for rolling partially immersed floating body, $BM = {}^{1}/_{V}$ where I is the second moment of area of the surface of floatation about the longitudinal axis, and V is immersed volume. (12 marks)

   (b) A cylindrical buoy weighing 20KN is to float in sea water above density is 1025kg/m^3. The buoy has a diameter of 2m and 2.5m high. Prove that it is unstable. (8 marks)

4. (a) Define the following terms as applied in fluid flow analysis;

   i. Eulerian and Langrarian methods of fluid flow (4 marks)
   ii. Velocity function (3 marks)
   iii. Stream function (3 marks)
   iv. Laminar and turbulent flow (4 marks)

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UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING/ BACHELOR OF SCIENCE IN AGRICULTURAL AND BIOS STEMS ENGINEERING

ECV 202/ EBE 202: FLUID MECHANICS I

DATE: FRIDAY 15TH MARCH 2019 **TIME:** 2.00 P.M. – 4.00 P.M.

K.V (ν) = $\frac{\text{Dynamic viscosity}}{\text{density } (\rho)}$

$\boxed{\nu = \frac{\mu}{\rho}}$

INSTRUCTIONS

I. Attempt QUESTION ONE and any other Two questions
II. Reasonable assumptions may be made where necessary

viscous force to inertia

Question 1

(a) Show that kinematic viscosity has primary dimension L^2T^{-1}. (3Marks) $\Rightarrow m^2/s$

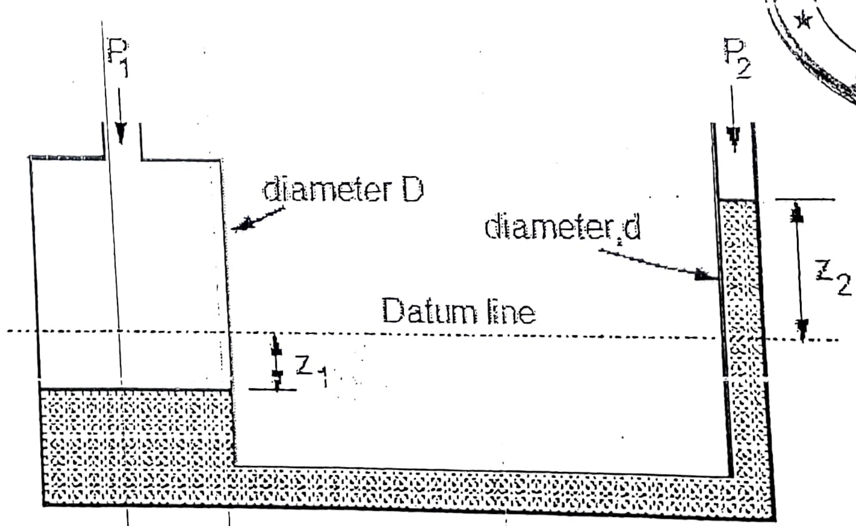
absolute viscosity / density

(b) State the Bernoulli's equation and the four assumptions governing its application. (6 Marks)

(c) State the six forces influencing fluid motion. (6 Marks)

(d) For the U-tube manometer shown below, prove that the difference in pressure is given by:

$$P_1 - P_2 = \rho g Z_2\left[1 + \left(\frac{d}{D}\right)^2\right]$$



Note → viscosity inc temp. of ga decrease of liquids

(5Marks)

(e) How deep can a diver descend in the ocean water without damaging his watch, which will withstand an absolute pressure of 5.5bar? Take the density of ocean water, $\rho = 1025 kg/m^3$ (4Marks)

(f) Using a relevant sketch demonstrate that the actual discharge through an orifice is $Q_{Act} = AC_d\sqrt{2gh}$ (6 marks)

Question 2

(a) A hydraulic jack having a ram 200 mm in diameter lifts a weight W = 40 kN under the action of a 25 mm plunger. What force is required on the plunger to lift the weight? (5Marks)

(b) A flat rectangular plate, dimension 1.5m length by 0.5m width is immersed in saline water (density 1025kg/m^3) such that their greatest and least depths are 1.50m and 0.60m respectively. Determine the force exerted on one face by the water pressure. (9Marks)

(c) In a fluid the velocity measured at a distance of 75mm from the boundary is 1.125m/s. the fluid has absolute viscosity 0.05 Pa s and relative density 0.93. What is the velocity gradient and shear stress at the boundary assuming a linear velocity distribution? Calculate its kinematic viscosity. (6 Marks)

Question 3

(a) Using a sketch and the relevant equations, show that the buoyancy or upthrust force on a floating body, $F = \rho g V_i$ (5 Marks)

(b) A rectangular pontoon has a width of 6m, a length L of 12m and a draught D of 1.5m in fresh water. Calculate

(i) The weight of the pontoon

(ii) The draught in sea water $\rho = 1025 kg/m^3$

(iii) The max load that can be carried by the pontoon in sea water if the max allowable draught is 2.0m (15 Marks)

ECU 202 15th March 2019.

Question 4

(a) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 4mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be unity. Density of air = 1.2KG/M^3. (10 marks)

(b) A venturi meter having an entrance diameter of 300mm and a throat diameter of 200mm is used to measure the volume of oil flowing through a pipe. The coefficient of discharge Cd is 0.95. If the density of oil ~~and~~ is equal to 0.9kg/m^3. Making reasonable assumptions, calculate the volume flow per second when the pressure difference between the entrance and the throat as measured on a water U-tube is 63mm. (10 Marks)

Question 5

(a) A tank has a circular orifice 20mm diameter in the vertical side near the bottom. It contains water to a depth of 1m above the orifice with oil of relative density 0.8 for a depth of 1m above the water. Acting on the upper surface of the oil is an air pressure on 20kN/m^2. The jet of water issuing from the orifice travels a horizontal distance of 1.5m from the orifice while falling a vertical distance of 0.156m. If the coefficient of contraction of the orifice is 0.65, estimate. find:

(i) The value of the coefficient of velocity, and

(ii) The actual discharge through the orifice. (10 Marks)

(b) Water flow over a rectangular notch and then over a 120^0 vee notch. The rectangular notch is 700mm long and the depth of water is 120mm. Find the corresponding depth over the V-notch. Assume coefficients of discharge to be 0.62 and 0.60 for the rectangular and V-notch respectively. (10 Marks)

ECV 202/EBE 202 15/03/2019

Question 4

(a) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 4mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be unity. Density of air = 1.2KG/M^3. (10 marks)

(b) A venturi meter having an entrance diameter of 300mm and a throat diameter of 200mm is used to measure the volume of oil flowing through a pipe. The coefficient of discharge Cd is 0.95. If the density of oil and is equal to ~~0.9kg/m^3~~ 900 kg/m^3. Making reasonable assumptions, calculate the volume flow per second when the pressure difference between the entrance and the throat as measured on a water U-tube is 63mm. (10 Marks)

Question 5

(a) A tank has a circular orifice 20mm diameter in the vertical side near the bottom. It contains water to a depth of 1m above the orifice with oil of relative density 0.8 for a depth of 1m above the water. Acting on the upper surface of the oil is an air pressure on 20kN/m^2. The jet of water issuing from the orifice travels a horizontal distance of 1.5m from the orifice while falling a vertical distance of 0.156m. If the coefficient of contraction of the orifice is 0.65, estimate. find:

(i) The value of the coefficient of velocity, and

(ii) The actual discharge through the orifice. (10 Marks)

(b) Water flow over a rectangular notch and then over a 120^0 vee notch. The rectangular notch is 700mm long and the depth of water is 120mm. Find the corresponding depth over the V-notch. Assume coefficients of discharge to be 0.62 and 0.60 for the rectangular and V-notch respectively. (10 Marks)

UNIVERSITY EXAMINATIONS 2014/2015

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECV 202: FLUID MECHANICS I

DATE: THURSDAY 11TH DECEMBER, 2014 TIME: 11.00 A.M – 1.00 P.M

INSTRUCTIONS:

- *Answer question 1 and any other TWO questions.*

1. (a) (i) At a certain point in castor oil the shear stress is 0.216N/m^2 and the velocity gradient is 0.216s^{-1}. If the mass density of castor oil is 959.42 kg/m^3. Find the kinematic viscosity (2 marks)

(ii) If 6m^3 of oil weighs 47KN, calculate the specific weight, mass density and specific gravity. (3 marks)

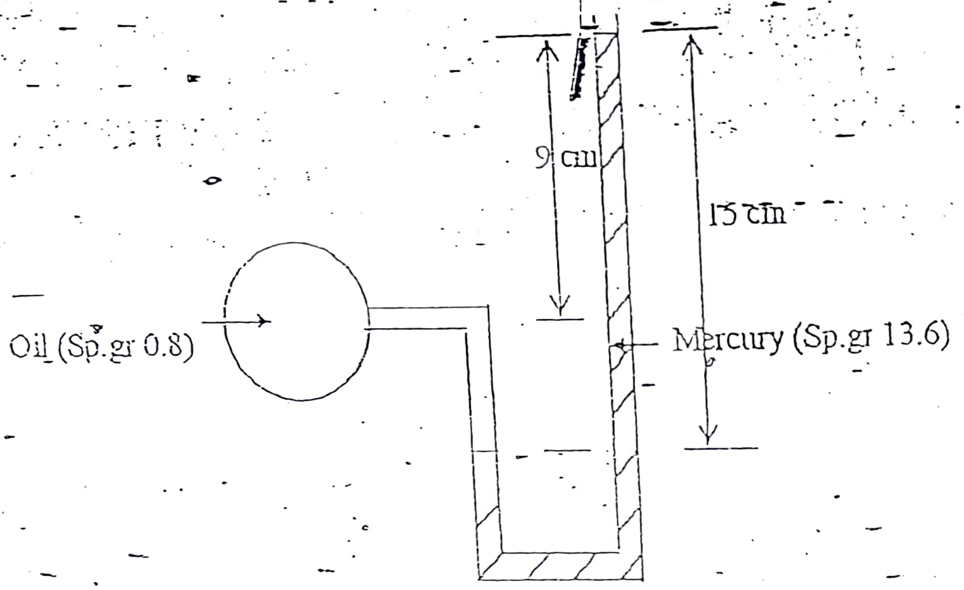
(iii) A clean tube of internal diameter 3 mm is immersed in a liquid with coefficient of surface tension 0.48 N/m. The angle of contact of the liquid with the glass can be assumed to be 130^0. The density of the liquid is 13600kg/m^3. What would be the level of the liquid in the tube relative to the free surface of the liquid outside the tube? (3 marks)

(iv) What is the bulk modulus of elasticity of a liquid which is compressed in a cylinder from volume of 0.0125 m^3 at 80N/cm^2 to a volume of 0.0124 m^3 at 150N/cm^2. (2 marks)

(b) Show that the intensity of pressure at a point in a fluid at rest is the same in all directions (6 marks)

(c) A square surface 3 m x 3m lies in a vertical plane. Determine the position of the centre of pressure and the total force on the square, when its upper edge is (a) in water surface and (b) 15m below the water surface. (4 marks)

(d) A U tube manometer is used to measure the pressure of oil (sp. gr 0.8) flowing in a pipeline. Its right limb is open to the atmosphere and the left limb is connected to the pipe. The centre of the pipe is 9cm below the level of mercury (sp. gr 13.6) in the right limb. If the difference of mercury level in the two limbs is 15cm, determine the absolute pressure of the oil in the pipe in KPa. (8 marks)



2. (a) A plane surface of area A is totally immersed in a liquid of specific weighγ. If the surface is inclined at an angle θ to the horizontal and its centroid is at a vertical depth $\bar{y}$ below the free surface, derive an expression for the resultant force F acting on the inclined plane surface. (9 marks)

(b) A 3.6m by 1.5m wide rectangular gate MN is vertical and is hinged at appoint 15cm below the centre of gravity of the gate. The total depth of water is 6m. What horizontal force must be applied at the bottom of the gate to keep the gate gate closed. (5 marks)

(c) A triangular gate which has a base of 1.5m and an altitude of 2m lies in a vertical plane. The vertex of the gate is 1m below the surface of a tank which contains oil of specific gravity 0.8. Find the force exerted by the oil on the gate and the position of the centre of pressure. (6 marks)

3. (a) Show that if B is the centre of buoyancy and M is the metacentre for rolling partially immersed floating body, $BM = {}^{I}/_{V}$ where I is the second moment of area of the surface of floatation about the longitudinal axis, and V is immersed volume. (12 marks)

(b) A cylindrical buoy weighing 20KN is to float in sea water above density is 1025kg/m^3. The buoy has a diameter of 2m and 2.5m high. Prove that it is unstable. (8 marks)

4. (a) Define the following terms as applied in fluid flow analysis;

i. Eulerian and Langranian methods of fluid flow (4 marks)

ii. Velocity function (3 marks)

iii. Stream function (3 marks)

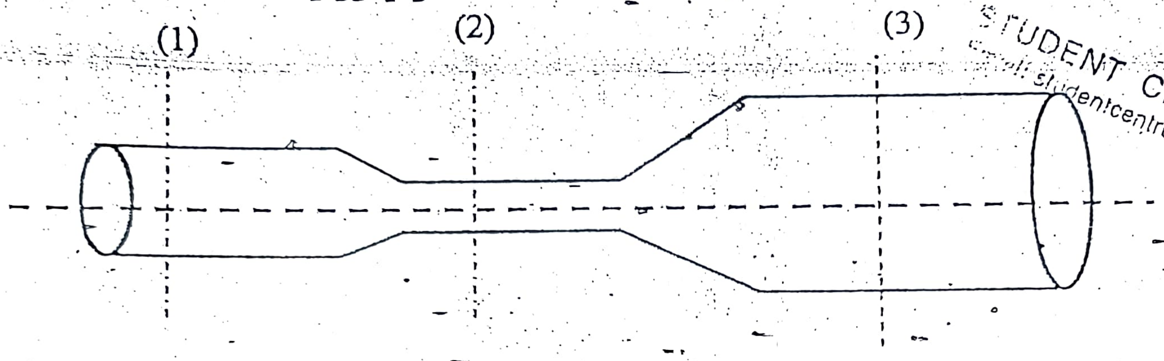
iv. Laminar and turbulent flow (4 marks)

ECV 202 11/12/2014

(b) If for two dimensional flow, the stream function is given by $\psi = y(3x^2 - y^2)$, obtain the velocity components and velocity at point (2,1). (8 marks)

5. (a) State the Bernoulli's theorem for a liquid (3 marks)

(b) An incompressible fluid flows through the circular pipe as shown in the figure below at the rate of $Q m^3/s$, it is assumed that the velocities at section (1), (2) and (3) are uniform, what are the velocities given that the diameter of pipe at three sections are $d_1 = 0.4m$, $d_2 = 0.2m$ and $d_3 = 0.75m$ respectively and Q = 0.3m3/s. (7 marks)



(a) A vessel containing a liquid of mass density 930kg/m^3 is given a constant vertical acceleration $f = 4.8m/s^2$ in an upward direction. If the vessel is 1.2m wide and 1.5m in breadth and the depth of the liquid is 0.9m, calculate the force on the bottom of the vessel;

i. While it is being accelerated (6 marks)

ii. When the acceleration ceases and the vessel continues to move at a constant velocity of 6m/s vertically upward. (4 marks)

- CLASS OF 2017

KENYATTA UNIVERSITY

CONTINUOUS ASSESSMENT TEST 2018/2019

BACHELOR OF SCIENCE IN CIVIL/ AGRICULTURAL & BIOSYSTEMS ENGINEERING

ECV 202 / EBE 202: FLUID MECHANICS I

(i) ATTEMPT A TOTAL OF FOUR QUESTIONS

(ii) TWO QUESTIONS SHOULD BE FROM PART I AND TWO FROM PART II

DATE: 27/02/2019

PART I

Question 1

(a) Derive the theoretical flow equation for a pitot static tube.

(b) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 3.0mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be 0.95. Density of air = 1.2Kg/m^3.

Question 2

(a) State the Pascal Law and using an example cite its engineering application.

(b) A venturi meter having a throat 110mm in diameter is fitted in a pipeline 220mm in diameter through which oil of specific gravity 0.9 is flowing at the rate of 0.15m^3/s. The inlet and throat of the meter are connected differentially to a U-tube manometer containing mercury of specific gravity 13.6 with oil immediately above this. Starting from Bernoulli's equation find the coefficient of discharge for the meter if the difference in mercury levels is 0.63m.

Question 3

(a) A tank 2.8 m high standing on the ground is kept full of water. There is an orifice in its vertical side at a depth hm below the surface of water. Find the value of h in order that the jet may strike the ground at a maximum distance from the tank.

KENYATTA UNIVERSITY

CONTINUOUS ASSESSMENT TEST 2018/2019

BACHELOR OF SCIENCE IN CIVIL/ AGRICULTURAL & BIOSYSTEMS ENGINEERING

ECV 202 / EBE 202: FLUID MECHANICS I

Date: 23rd Jan 2019: 7.30 - 9.30am

(i) ATTEMPT ALL THREE QUESTIONS

(ii) REASONABLE ASSUMPTIONS MAY BE MADE WHERE NECESSARY.

Question 1

(a) What is the definition of fluid in fluid mechanics

(b) A hydraulic jack having a ram 250 mm in diameter lifts a weight W = 40 kN under the action of a 35 mm plunger. What force is required on the plunger to lift the weight?

(c) Show that dynamic viscosity has primary dimension $ML^{-1}T^{-1}$.

(d) A pitot-static tube is used to measure air velocity. If a manometer connected to the instrument indicates a difference in pressure head between the tappings of 5mm of water, calculate the air velocity assuming the coefficient of the pitot tube to be 0.98. Density of air = 1.2Kg/m^3.

Question 2

(a) Using a simple sketch state and proof the Parallel Axis Theorem

(b) A flat rectangular plate, dimension 1.5m length by 0.5m width is immersed in saline water (density 1025kg/m^3) such that their greatest and least depths are 1.80m and 0.80m respectively. Determine the force exerted on one face by the water pressure.

Question 3

Crude oil of density 850Kg/m^3 flows upwards at a volume rate of 0.06 m^3/s through a vertical venturi meter which has an inlet diameter of 200mm and a throat diameter 100mm. The venturi coefficient is 0.9. The vertical distance between the pressure tappings is 300mm and they are connected:

(i) To two pressure gauges calibrated in kN/m^2 and positioned at the levels of their corresponding tapping points.

(ii) To the two limbs of a mercury manometer (ρ_{Hg} = 13600Kg/m^3), the connecting tube being filled with oil.

Determine the difference of the readings of 2 pressure gauges and difference of the level of mercury column in the manometer.